

Harsh Desai

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EDUCATION

University of Illinois Chicago
B.S. Computer Science

Expected May 2027

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, HTML/CSS, Swift, SQL, Lua, C#, Qiskit

Frameworks & Tools: React, Next.js, Spring Boot, MongoDB, Git/GitHub, Jupyter Notebook, Qbraid, Xcode, Unity, IntelliJ, VS Code, AWS

EXPERIENCE

Quantum Algorithms Researcher | University of Illinois Chicago

Aug 2025 – Mar 2026

- Investigated Jordan-Wigner vs. Bravyi-Kitaev fermion-to-qubit mappings to reduce circuit complexity by 75%, benchmarking simulation efficiency across multi-qubit systems while maintaining equivalent time-control accuracy.
- Designed experimental frameworks in Qiskit to benchmark qubit mapping approaches, producing reproducible comparisons across multi-qubit quantum circuit configurations.

Quantum Software Development Intern | University of Illinois Chicago

Summer 2025

- Doubled measurement accuracy (60%) across variable configurations by building Python scripts to compute expectation values for Quantum Harmonic Oscillators and multi-qubit systems.
- Produced reproducible experimental results analyzing behavioral shifts in Quantum Harmonic Oscillators under both incremental and drastic variable changes.

Computer Science Instructor | Code Ninjas, Long Grove

2022 – 2024

- Delivered a full-stack development lifecycle curriculum for 8+ students aged 8–13, covering programming fundamentals, game design, and AI concepts using Roblox Studio, Scratch, and Code Combat.
- Created and iterated custom lesson plans and slide decks for rotating class cohorts, adapting material for varying skill levels based on student feedback.

PROJECTS

CurtainCall | *Next.js, React, Spring Boot, MongoDB, Tailwind CSS*

[Live Demo](#) · *Feb 2026*

- Engineered a Tinder-style artist networking platform with a swipe-based matching system, shipping 12+ RESTful API endpoints in Spring Boot to support user connections, endorsements, and discovery.
- Architected MongoDB collections around user relationships and endorsement logic, structuring schemas to support 1,000+ concurrent users while eliminating redundant database calls.

SnapLink | *React, Spring Boot, MongoDB, Google Maps API*

[Live Demo](#) · *Feb 2025*

- Developed a geo-tagged task discovery platform connecting small businesses with local job-seekers in real time, integrating the Google Maps API with dynamic geolocation filtering for hyper-local opportunity matching.
- Implemented a full-stack architecture with RESTful Spring Boot APIs and MongoDB, enabling scalable task posting, status tracking, and location-based querying.

Lucky D's Gaming Cafe Website | *React*

[Live Site](#) · *Jun 2022*

- Independently designed, built, and deployed a production React site for a local gaming café, owning the full SDLC from requirements through deployment and post-launch maintenance. Hosted from Wix.
- Has served real customer traffic continuously since launch (approx: 5000 users), letting visitors browse live events and services before visiting in person.

CountingCards | *HTML, CSS, JavaScript, Bootstrap*

[Public Link](#) · *Feb 2026*

- Created a gamified card-based study system with dynamic card cycling, interactive session tracking, and a responsive Bootstrap UI, replacing passive reading with active recall loops.

AWARDS & LEADERSHIP

- IBM Quantum Badge of Excellence — completed all labs in the Qiskit Global Summer School 2025 with 100% accuracy.
- Freshman Mentor Program — selected as 1 of 100 mentors from 2,000+ applicants; recognized as a top mentor by program leadership.
- IEEE Executive Board — Web development board member; leading redesign and modernization of the IEEE UIC chapter website.